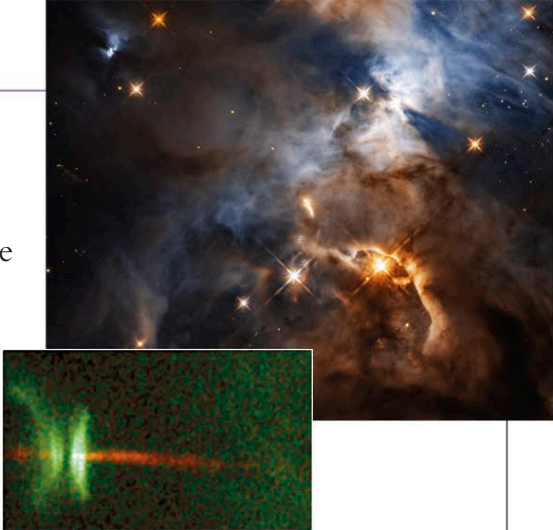


EXTRASOLAR PLANETS

25	Stars and brown dwarfs
90–91	Astronomical observatories
94–95	Observing from space
235	Formation of a planetary system
238–239	Star formation
274–275	Multiple stars

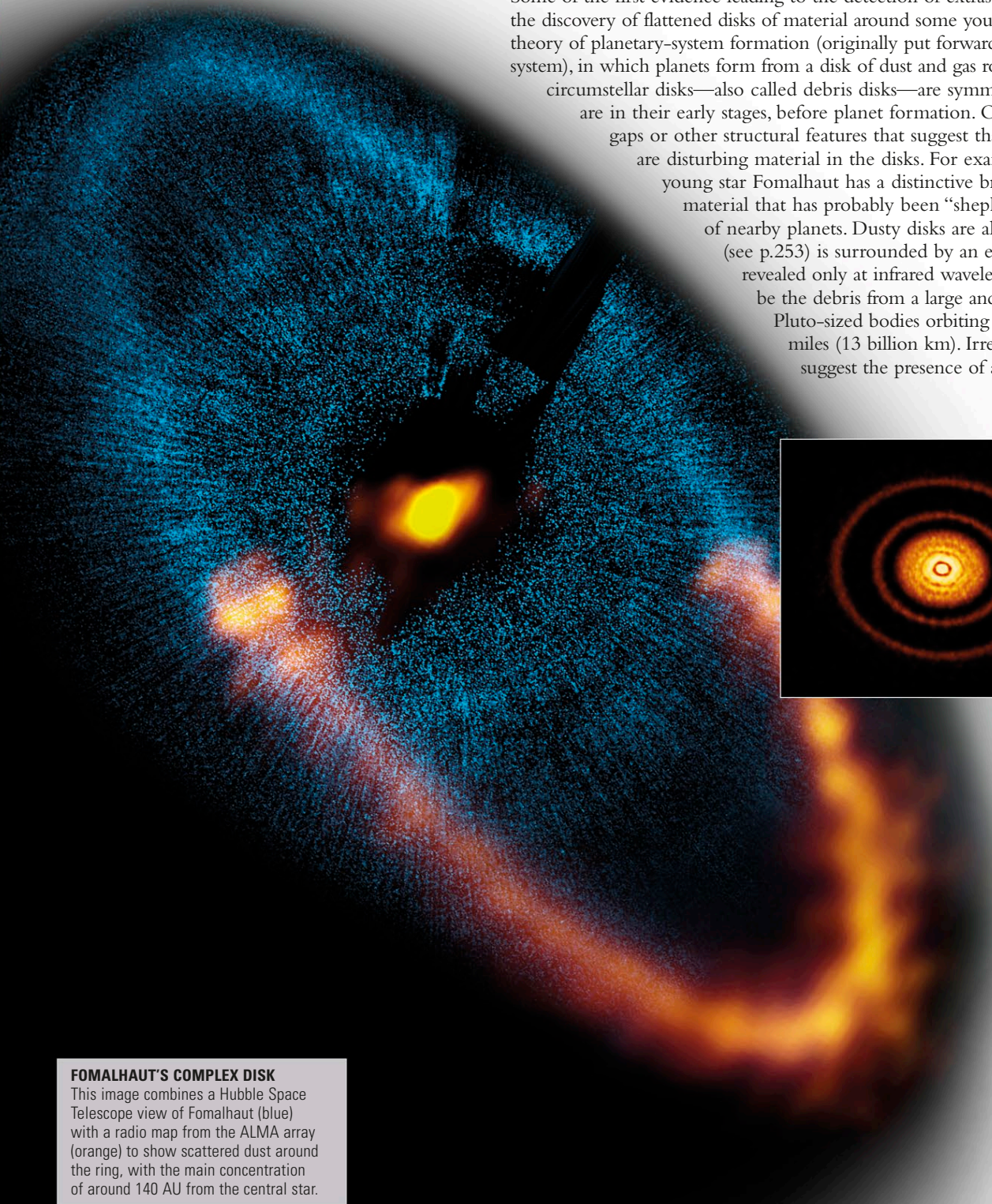
THE SUN IS NOT the only star with a planetary system. More than 4,000 planets have so far been found orbiting other stars, with the list growing rapidly every year. Extrasolar planets have been detected around stars of a range of types and ages, suggesting that planet formation is a robust process and that planetary systems are commonplace.

TELLTALE SHADOWS
Planet-forming disks around young stars can be seen in direct silhouette (right) or when they cast twin lighthouse-like shadows across nearby nebulae (top right).

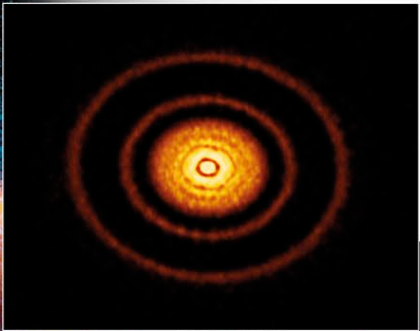


PLANET-FORMING DISKS

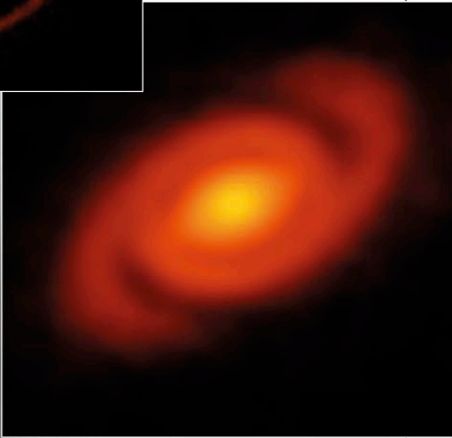
Some of the first evidence leading to the detection of extrasolar planets, or exoplanets, was the discovery of flattened disks of material around some young stars. This fit the standard theory of planetary-system formation (originally put forward to explain the origins of the solar system), in which planets form from a disk of dust and gas rotating around a star. Some such circumstellar disks—also called debris disks—are symmetrical, suggesting that they are in their early stages, before planet formation. Others are distorted or have gaps or other structural features that suggest that planets have formed and are disturbing material in the disks. For example, the disk around the bright young star Fomalhaut has a distinctive bright outer ring of concentrated material that has probably been “shepherded” into place by the influence of nearby planets. Dusty disks are also found around mature stars. Vega (see p.253) is surrounded by an extensive dust disk, which is fully revealed only at infrared wavelengths. This fine dust is thought to be the debris from a large and relatively recent collision between Pluto-sized bodies orbiting the star at a distance of 8 billion miles (13 billion km). Irregularities in Vega’s debris disk also suggest the presence of at least one planet.



FOMALHAUT’S COMPLEX DISK
This image combines a Hubble Space Telescope view of Fomalhaut (blue) with a radio map from the ALMA array (orange) to show scattered dust around the ring, with the main concentration of around 140 AU from the central star.



YOUNG DISK
The protoplanetary disk around the star AS 209 appears perfect—the star is so young that no planets have yet had time to form.



DEVELOPING SPIRAL
The debris disk around the young star Elias 2-27 has formed a spiral pattern with concentrations of matter out of which planets may soon emerge. The structure is created by density waves similar to those seen in spiral galaxies (see p.303).